

ASSIGNMENT-6

Subject – Computer Science Workshop-1(CSE 2141)

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/*Q1. Write a program to convert an integer to an Integer object.
(a) Autoboxing (b) Using Constructor

JAVA CODE

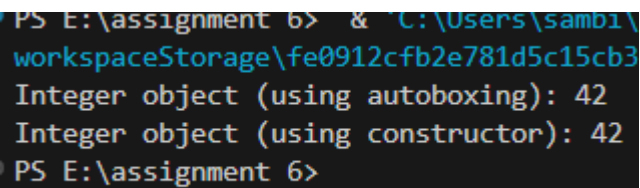
```
package mypackage;

public class ASS6_Q1 {
    public static void main(String[] args) {
        int num = 42;

        Integer integerAutoboxed =num;
        System.out.println("Integer object (using autoboxing): " + integerAutoboxed);

        Integer integerUsingConstructor = new Integer(num);
        System.out.println("Integer object (using constructor): " +
integerUsingConstructor);
    }
}
```

OUTPUT



```
PS E:\assignment 6> java -cp . ASS6_Q1
Integer object (using autoboxing): 42
Integer object (using constructor): 42
PS E:\assignment 6>
```

Q2. Write a program to convert a float to a Float object. (a) Autoboxing (b) Using Constructor

JAVA CODE :-

```
package mypackage;

public class ASS6_Q2 {
    public static void main(String[] args) {
        float num = 42;

        Float floatAutoboxed = num;
        System.out.println("Float object (using autoboxing): " + floatAutoboxed);

        @SuppressWarnings("removal")
        Float floatUsingConstructor = new Float(num);
        System.out.println("Integer object (using constructor): " +
floatUsingConstructor);
    }
}
```

OUTPUT :-

```
PS E:\workspace\storage (10091201020701051)
Float object (using autoboxing): 42.0
Integer object (using constructor): 42.0
PS E:\assignment 6>
```

Q3. Write a program to convert a double to a Double object. (a) Autoboxing (b) Using Constructor

JAVA CODE :-

```
package mypackage;

public class ASS6_Q3 {
    public static void main(String[] args) {
        Double num = 42.0d;

        Double doubleAutoboxed = num;
        System.out.println("Double object (using autoboxing): " + doubleAutoboxed);

        @SuppressWarnings("removal")
        Double doubleUsingConstructor = new Double(num);
        System.out.println("Double object (using constructor): " +
doubleUsingConstructor);
    }
}
```

OUTPUT:-

```
Double object (using autoboxing): 42.0
Double object (using constructor): 42.0
PS E:\assignment 6>
```

Q4. Write a program to convert a boolean to a Boolean object. (a) Autoboxing (b) Using Constructor

JAVA CODE

```
package mypackage;

public class ASS6_Q4 {
    public static void main(String[] args) {
        boolean num = true;

        Boolean booleanAutoboxed =num;
        System.out.println("Boolean object (using autoboxing): " + booleanAutoboxed);

        @SuppressWarnings("removal")
        Boolean booleanUsingConstructor = new Boolean(num);
        System.out.println("Boolean object (using constructor): " +
booleanUsingConstructor);
    }
}
```

OUTPUT :-

```
Boolean object (using autoboxing): true
Boolean object (using constructor): true
PS: 5\assignment 6\
```

Q5. Write a program to read an integer as a string and convert it to an Integer object.

JAVA CODE

```
package mypackage;
```

```
public class ASS6_Q5 {  
    public static void main(String args[]) {  
        String s = "10";  
  
        Integer integerObject = Integer.valueOf(s);  
        System.out.println("String \"\" + s + \"\" converted to Integer object is: \" +  
integerObject);  
    }  
}
```

OUTPUT :-

```
String "10" converted to Integer object is: 10
```

Q6. Write a program to read a float as a string and convert it to a Float object.

JAVA CODE :-

```
package mypackage;
import java.util.*;
public class ASS6_Q6 {
    public static void main(String args[]) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a string to convert it to ");
        String s = sc.next();

        Float floatObject = Float.parseFloat(s);
        System.out.println("String \"\" + s + \"\" converted to Float object is: " +floatObject);
    }
}
```

OUTPUT :-

```
Enter a string to convert it to
45.36
String "45.36" converted to Float object is: 45.36
PS E:\assignment 6> █
```

Q7. Write a program to read a double as a string and convert it to a Double object.

JAVA CODE

```
package mypackage;
import java.util.*;
public class Dec16 ASS6_Q7 {
    public static void main(String args[]) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a string to convert it to ");
        String s = sc.next();

        Double doubleObject = Double.parseDouble(s);
        System.out.println("String \"\" + s + \"\" converted to Double object is: "
+doubleObject);
    }
}
```

OUTPUT :-

```
Enter a string to convert it to
45.63
String "45.63" converted to Double object is: 45.63
PS E:\assignment 6> █
```


Q8. Write a program to read a boolean as a string and convert it to a Boolean object. Explain the concept of converting a base data type to an object type(Wrapping) using the valueOf() method.

JAVA CODE :-

```
package mypackage;
import java.util.*;
public class ASS6_Q8 {
    public static void main(String args[]) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a string to convert it to ");
        String s = sc.next();

        Boolean booleanObject = Boolean.parseBoolean(s);
        System.out.println("String \"\" + s + \"\" converted to Boolean object is: "
+booleanObject);
    }
}
```

OUTPUT :-

```
Enter a string to convert it to
true
String "true" converted to Boolean object is: true
PS E:\assignment 6> █
```

Q9. Write a program that reads to convert int, float, double, and boolean as string types and convert them to respective object types using the valueOf method.

JAVA CODE :-

```
package mypackage;
import java.util.*;
public class ASS6_Q9 {
    public static void main(String args[]) {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter a string to convert it to integer");
        String si = sc.next();
        System.out.println("Enter a string to convert it to float");
        String sf = sc.next();
        System.out.println("Enter a string to convert it to double");
        String sd = sc.next();
        System.out.println("Enter a string to convert it to boolean");
        String sb = sc.next();

        Integer integerObject = Integer.valueOf(si);
        Float floatObject = Float.valueOf(sf);
        Double doubleObject = Double.valueOf(sd);
        Boolean booleanObject = Boolean.valueOf(sb);
        System.out.println("String \"\" + si + \"\" converted to Integer object is: \" +
integerObject);
        System.out.println("String \"\" + sf + \"\" converted to Float object is: \" + floatObject);
        System.out.println("String \"\" + sd + \"\" converted to Double object is: \" +
doubleObject);
        System.out.println("String \"\" + sb + \"\" converted to Boolean object is: \" +
booleanObject);
    }
}
```

OUTPUT :-

```
Enter a string to convert it to integer
45
Enter a string to convert it to float
45.306
Enter a string to convert it to double
45.966
Enter a string to convert it to boolean
true
String "45" converted to Integer object is: 45
String "45.306" converted to Float object is: 45.306
String "45.966" converted to Double object is: 45.966
String "true" converted to Boolean object is: true
```

Q10. Write a program to design a simple calculator (only +,-,*,/ operations). The calculator works as follows: Input: "123+345" Output: Sum=468 Input: "5*10" Output: mul=50 Explain the concept of converting object type to base type. Explain the method used to do so.

```
package mypackage;

import java.util.Scanner;

public class ASS6_Q10{
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Take user input in the format "number1operatornumber2"
        System.out.print("Enter an operation (e.g., 123+345): ");
        String input = scanner.nextLine();

        // Variables to store parsed values
        int num1 = 0, num2 = 0;
        char operator = '\0';

        // Parse the input to separate the numbers and the operator
        for (int i = 0; i < input.length(); i++) {
            if (input.charAt(i) == '+' || input.charAt(i) == '-' || input.charAt(i) == '*' ||
input.charAt(i) == '/') {
                operator = input.charAt(i);
                num1 = Integer.parseInt(input.substring(0, i)); // Convert first part to integer
                num2 = Integer.parseInt(input.substring(i + 1)); // Convert second part to
integer
                break;
            }
        }

        // Perform the operation and display the result
        switch (operator) {
            case '+':
                System.out.println("Sum=" + (num1 + num2));
                break;
            case '-':
                System.out.println("Sub=" + (num1 - num2));
                break;
            case '*':
                System.out.println("Mul=" + (num1 * num2));
                break;
            case '/':
                if (num2 != 0) {
                    System.out.println("Div=" + (num1 / num2));
                } else {
                    System.out.println("Error: Division by zero");
                }
                break;
        }
    }
}
```

```

        default:
            System.out.println("Invalid operator");
        }

        scanner.close();
    }
}

```

EXPLANATION OF THE concept of converting object type to base type. Explain the method used to do so.

/*Here is a Java program for a simple calculator that can handle addition (+), subtraction (-), multiplication (*), and division (/).
The program uses basic string manipulation to parse the input and performs the requested operation.
Following the program, an explanation of converting object types to base types is provided.

Explanation of Converting Object Types to Base Types in Java

In Java, primitive data types (like `int`, `float`, etc.) and their corresponding wrapper classes (like `Integer`, `Float`, etc.) are distinct.

Wrapper classes are object representations of the primitive types. Conversion between these types is done using:

1. ****Unboxing****: This refers to converting an object type (like `Integer`) into its base type (`int`). It happens automatically in Java because of ****autoboxing and unboxing**** introduced in Java 5.

- Example:

```

```java
Integer obj = Integer.valueOf(42); // Wrapper class object
int baseType = obj; // Automatic unboxing to primitive
```

```

2. ****Explicit Parsing****: When working with `String` inputs, you often need to convert these to base types. This is done using methods like `Integer.parseInt(String)` or `Double.parseDouble(String)`.

- Example from the program:

```

```java
num1 = Integer.parseInt(input.substring(0, i)); // Converts String to int
```

```

Method Used in the Program

The program uses the `Integer.parseInt(String)` method to explicitly convert the string representation of a number into its primitive type (`int`). This is necessary because user input from the console is always read as a `String`. The method ensures that mathematical operations can be performed on the numeric values. */

Q11. Write a program that reads a double number as a sting and converts it to a double base type.

JAVA CODE :-

```
package mypackage;
import java.util.Scanner;

public class ASS6_Q11 {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a double value as a string: ");
        String str = scanner.nextLine();

        try {
            double doubleValue = Double.parseDouble(str);
            System.out.println("The string \"" + str + "\" converted to a double is: " +
doubleValue);
        } catch (NumberFormatException e) {
            System.out.println("Error: Invalid input. Please enter a valid double value.");
        }
        scanner.close();
    }
}
```

OUTPUT : -

```
Enter a double value as a string: 45.3602
The string "45.3602" converted to a double is: 45.3602
PS E:\assignment 6>
```

Q12. Write a program that reads an integer number as a sting and converts it to an int base type. Explain the following concepts:

- Arrays
- Conditional Statements
- Loops

JAVA CODE :-

```
package mypackage;
import java.util.Scanner;

public class ASS6_Q12 {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter a integer value as a string: ");
        String str = scanner.nextLine();

        try {
            int integerValue = Integer.parseInt(str);
            System.out.println("The string \"\" + str + "\" converted to a integer is: \" +
integerValue);
        } catch (NumberFormatException e) {
            System.out.println("Error: Invalid input. Please enter a valid integer value.");
        }
        scanner.close();
    }
}
```

EXPLANATION

Explanation of Concepts

Arrays

- **Definition:** Arrays are data structures that store multiple values of the same type in a single variable. Each element in an array is accessed using an index.
- **Example:**
- `int[] numbers = {1, 2, 3, 4};`
- `System.out.println(numbers[2]);` // Outputs 3

Conditional Statements

- **Definition:** Conditional statements allow the program to execute certain code blocks based on whether a condition is true or false.
- **Types:** if, else if, else, and switch.
- **Example:**
- `int number = 10;`
- `if (number > 5) {`
- `System.out.println(\"Number is greater than 5\");`
- `} else {`
- `System.out.println(\"Number is 5 or less\");`
- `}`

Loops

- **Definition:** Loops are used to repeatedly execute a block of code as long as a specified condition is true.

- **Types:** for, while, and do-while.
- **Example:**
- `for (int i = 0; i < 5; i++) {`
- `System.out.println("Iteration: \" + i);`
- `}`

Summary

The program converts a string representation of a number into its integer form using `Integer.parseInt`. It employs conditional statements (try-catch) to handle invalid input gracefully. Arrays, loops, and conditionals are core constructs in Java programming, enabling efficient data management and control flow.

OUTPUT :-

```
PS E:\assignment 02 > & "C:\Users\samb1\AppData\Local\Programs\Microsoft VS Code\bin\code.cmd" %*  
C:\Users\samb1\AppData\Local\Programs\Microsoft VS Code\bin\code.cmd  
Enter a integer value as a string: 10025  
The string "10025" converted to a integer is: 10025  
PS E:\assignment 6> █
```

Q13. Write a program that prompts the user to input a positive integer. It should then print the multiplication table of that number.

JAVA CODE

```
import java.util.Scanner;

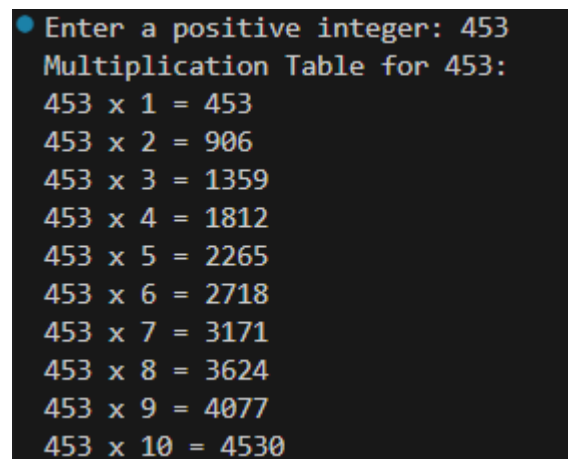
public class MultiplicationTable {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Prompt the user to enter a positive integer
        System.out.print("Enter a positive integer: ");
        int number = scanner.nextInt();

        // Validate the input
        if (number <= 0) {
            System.out.println("Error: Please enter a positive integer.");
        } else {
            // Print the multiplication table
            System.out.println("Multiplication Table for " + number + ":");
            for (int i = 1; i <= 10; i++) {
                System.out.println(number + " x " + i + " = " + (number * i));
            }
        }

        scanner.close();
    }
}
```

OUTPUT :-



```
● Enter a positive integer: 453
Multiplication Table for 453:
453 x 1 = 453
453 x 2 = 906
453 x 3 = 1359
453 x 4 = 1812
453 x 5 = 2265
453 x 6 = 2718
453 x 7 = 3171
453 x 8 = 3624
453 x 9 = 4077
453 x 10 = 4530
```


Q14. Write a java program to calculate HCF and LCM of Two given number.

JAVA CODE

```
import java.util.Scanner;
public class HCFAndLCM {
    public static int calculateHCF(int num1, int num2) {
        while (num2 != 0) {
            int temp = num2;
            num2 = num1 % num2;
            num1 = temp;
        }
        return num1;
    }
    public static int calculateLCM(int num1, int num2, int hcf) {
        return (num1 * num2) / hcf;
    }
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the first number: ");
        int num1 = scanner.nextInt();
        System.out.print("Enter the second number: ");
        int num2 = scanner.nextInt();
        int hcf = calculateHCF(num1, num2);
        int lcm = calculateLCM(num1, num2, hcf);
        System.out.println("HCF of " + num1 + " and " + num2 + " is: " + hcf);
        System.out.println("LCM of " + num1 + " and " + num2 + " is: " + lcm);
        scanner.close();
    }
}
```

OUTPUT

```
Enter the first number: 12
Enter the second number: 18
HCF of 12 and 18 is: 6
LCM of 12 and 18 is: 36
```

Q15 .Write a program to calculate the sum of following series where n is input by user. $1 + 1/2 + 1/3 + 1/4 + 1/5 + \dots + 1/n$

```
import java.util.Scanner;
public class SeriesSum {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input the value of n
        System.out.print("Enter the value of n: ");
        int n = scanner.nextInt();

        // Validate input
        if (n <= 0) {
            System.out.println("Please enter a positive integer.");
            return;
        }

        double sum = 0.0;

        // Calculate the sum of the series
        for (int i = 1; i <= n; i++) {
            sum += 1.0 / i;
        }

        // Output the result
        System.out.printf("The sum of the series is: %.6f\n", sum);

        scanner.close();
    }
}
```

OUTPUT :-

```
Enter the value of n: 452
The sum of the series is: 6.692004
```

Q16. Write a program to enter the numbers till the user wants and at the end the program should display the largest and smallest numbers entered.

JAVA CODE :-

```
import java.util.Scanner;

public class ASS6_Q16 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int largest = Integer.MIN_VALUE;
        int smallest = Integer.MAX_VALUE;
        char choice;

        do {
            System.out.print("Enter a number: ");
            int number = scanner.nextInt();

            // Update largest and smallest
            if (number > largest) {
                largest = number;
            }
            if (number < smallest) {
                smallest = number;
            }

            System.out.print("Do you want to enter another number? (y/n): ");
            choice = scanner.next().toLowerCase().charAt(0);

        } while (choice == 'y');

        // Display results
        System.out.println("Largest number entered: " + largest);
        System.out.println("Smallest number entered: " + smallest);

        scanner.close();
    }
}
```

OUTPUT :-

```
Enter a number: 45
Do you want to enter another number? (y/n): y
Enter a number: 21
Do you want to enter another number? (y/n): n
Largest number entered: 45
Smallest number entered: 21
```

Q17. . Write a java program to find the minimum and maximum element in an array.

JAVA CODE :-

```
import java.util.Scanner;

public class ASS6_Q17 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input: Array size
        System.out.print("Enter the size of the array: ");
        int size = scanner.nextInt();

        // Input: Array elements
        int[] array = new int[size];
        System.out.println("Enter " + size + " elements:");
        for (int i = 0; i < size; i++) {
            array[i] = scanner.nextInt();
        }

        // Initialize min and max
        int min = array[0];
        int max = array[0];

        // Traverse the array to find min and max
        for (int i = 1; i < size; i++) {
            if (array[i] < min) {
                min = array[i];
            }
            if (array[i] > max) {
                max = array[i];
            }
        }

        // Output the results
        System.out.println("Minimum element: " + min);
        System.out.println("Maximum element: " + max);

        scanner.close();
    }
}
```

OUTPUT

```
Enter the size of the array: 5
Enter 5 elements:
4 5 6 7 2
Minimum element: 2
Maximum element: 7
```

Q18. Write java program to find the Kth largest and Kth smallest number in an array.

JAVA CODE

```
import java.util.Arrays;
import java.util.Scanner;

public class ASS6_Q18 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input: Array size
        System.out.print("Enter the size of the array: ");
        int size = scanner.nextInt();

        // Input: Array elements
        int[] array = new int[size];
        System.out.println("Enter " + size + " elements:");
        for (int i = 0; i < size; i++) {
            array[i] = scanner.nextInt();
        }

        // Input: Value of K
        System.out.print("Enter the value of K: ");
        int k = scanner.nextInt();

        // Check if K is valid
        if (k > size || k <= 0) {
            System.out.println("Invalid value of K. It should be between 1 and " + size);
            return;
        }

        // Sort the array
        Arrays.sort(array);

        // Find K-th smallest and K-th largest
        int kthSmallest = array[k - 1];
        int kthLargest = array[size - k];

        // Output the results
        System.out.println("K-th smallest element: " + kthSmallest);
        System.out.println("K-th largest element: " + kthLargest);

        scanner.close();
    }
}
```

OUTPUT :-

```
Enter the size of the array: 10
Enter 10 elements:
4 5 6 7 5 8 4 5 6 9
Enter the value of K: 2
K-th smallest element: 4
K-th largest element: 8
```

Q19. Write a java program to reverse the given array. (Without using Library function)

JAVA CODE

```
import java.util.Scanner;

public class ReverseArrayWithWhileLoop {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input: Array size
        System.out.print("Enter the size of the array: ");
        int size = scanner.nextInt();

        // Input: Array elements
        int[] array = new int[size];
        System.out.println("Enter " + size + " elements:");
        for (int i = 0; i < size; i++) {
            array[i] = scanner.nextInt();
        }

        // Reverse the array using a while loop
        int left = 0;
        int right = size - 1;
        while (left < right) {
            // Swap elements at left and right indices
            int temp = array[left];
            array[left] = array[right];
            array[right] = temp;

            // Move the indices towards each other
            left++;
            right--;
        }

        // Output: Reversed array
        System.out.println("Reversed array:");
        for (int element : array) {
            System.out.print(element + " ");
        }

        scanner.close();
    }
}
```

OUTPUT :-

```
Enter the size of the array: 5
Enter 5 elements:
4 5 7 9 5
Reversed array:
5 9 7 5 4
```

Q20. Write a java program to sort the given array. (Without using Library function)

JAVA CODE :-

```
import java.util.Scanner;

public class ASS6_Q20 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input: Array size
        System.out.print("Enter the size of the array: ");
        int size = scanner.nextInt();

        // Input: Array elements
        int[] array = new int[size];
        System.out.println("Enter " + size + " elements:");
        for (int i = 0; i < size; i++) {
            array[i] = scanner.nextInt();
        }

        // Sorting the array using Bubble Sort
        for (int i = 0; i < size - 1; i++) {
            for (int j = 0; j < size - i - 1; j++) {
                if (array[j] > array[j + 1]) {
                    // Swap array[j] and array[j+1]
                    int temp = array[j];
                    array[j] = array[j + 1];
                    array[j + 1] = temp;
                }
            }
        }

        // Output: Sorted array
        System.out.println("Sorted array:");
        for (int element : array) {
            System.out.print(element + " ");
        }

        scanner.close();
    }
}
```

```
}
```

OUTPUT:-

```
Enter the size of the array: 10
Enter 10 elements:
2 9 8 7 5 1 2 9 8 9
Sorted array:
1 2 2 5 7 8 8 9 9 9
Enter the size of the array: 10
Enter 10 elements:
2 9 8 7 5 1 2 9 8 9
Sorted array:
1 2 2 5 7 8 8 9 9 9
```